

X̄ - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
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Revisi	00
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Tgl Berlaku	2-Dec-16
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MACHINE Friction Oil	SECTION QC	OPERATION Sliding	DATE CONTROL LIMITS CALC.		SPECIFICATION minmax 6.8		COURSE NO.	CONTROL ITEM (Δ) YES NO	O
	MONTH May-25	CHARACTERISTIC Kandungan Oil			SAMPLE SIZE / FREQUENCY n = 3		TYPE / SIZE : D2IDW4-B-2 / 0.70 PIC : HERIBERTUS		

$$\bar{X} = \text{AVERAGE } \bar{X} = 0.038 \quad \text{U.C.L.} = \bar{X} + A_2 \bar{R} = 0.0460 \quad \text{L.C.L.} = \bar{X} - A_2 \bar{R} = 0.0300$$

	SPECIAL CAUSES
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- ✓ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ✓ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
- ✓ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ✓ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ✓ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ✓ TEST 8: 9 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

1. _____
2. _____
3. _____
4. _____
5. _____

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

CALCULATION

$$\begin{aligned} \bar{X} &= 0.0367 \\ \bar{R} &= 0.0048 \\ s_{ST} &= 0.0028 & s_{LT} &= 0.003 \end{aligned}$$

THE PROCESS MUST BE IN CONTROL
BEFORE CAPABILITY CAN BE DETERMINED

$$C_{PK} = \min \left[\frac{\bar{X} - LSL}{3s_{ST}}, \frac{USL - \bar{X}}{3s_{ST}} \right]$$

$$= \min \left[0.79, 1.10 \right]$$

$$= 0.79$$
$$C_p = \left[\frac{USL - LSL}{6\sigma_{ST}} \right]$$

$$= \left[0.94 \right]$$

CHECKED BY	APPROVED BY
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